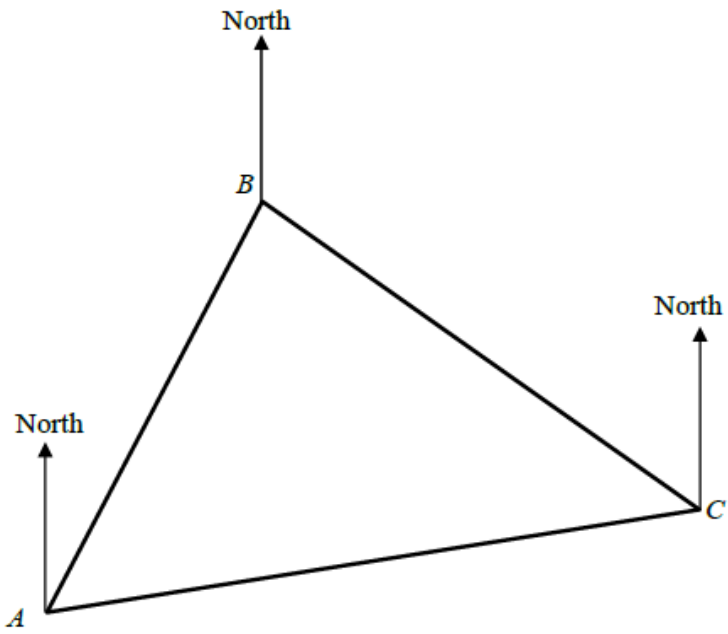
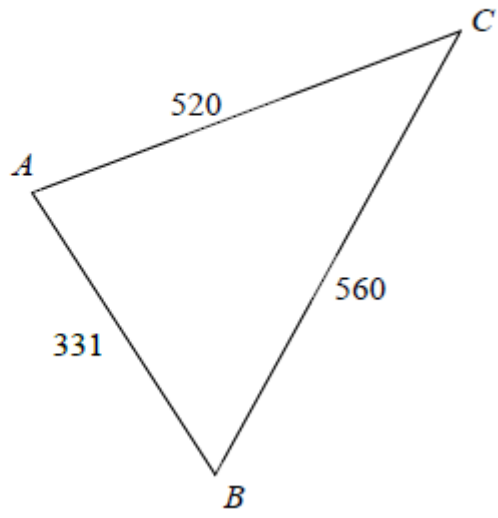
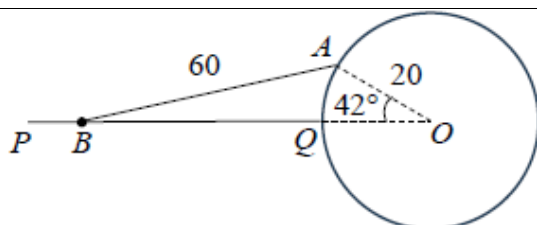


1	YTSS/2025/I/20
	The diagram below represents three ride stations A, B and C in a theme park.  (a) Find the bearing of B from A. [1] (b) Construct the perpendicular bisector of AC. [1] (c) Construct the bisector of angle ACB. [1] (d) A restroom, R, is to be built inside the theme park so that it is closer to A than to C and closer to AC than to BC. Shade the region where the restroom is to be built. [1]
2	YTSS/2025/I/22
	The diagram below shows the locations of A, B and C. It is given that $AB = 331$ m, $AC = 520$ m and $BC = 560$ m.  (a) Find angle CAB in degrees. [1]

- (b) A building, D , is to be built along BC such that it gives the greatest angle of elevation from A to top of building, T , which is 18.4° .
Calculate the vertical height of the building. [4]

3 SCGS/2025/I/20

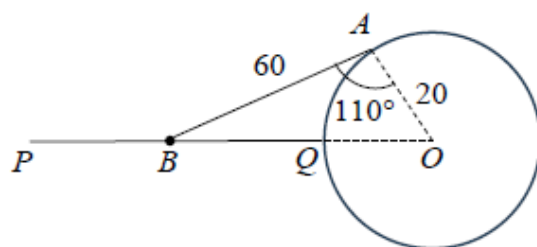


In the diagram, A and Q are points on the rim of a circular wheel, centre O .
 PQ is a fixed wire such that PQO is a straight line.
A string of length 60 cm has one end attached to A and the other end attached to a bead, B , that is threaded on PQ .
 $OA = 20$ cm and angle $AOQ = 42^\circ$.

- (a) Calculate angle ABQ .

Answer Angle $ABQ = \dots\dots\dots$ [2]

The wheel is slowly rotated clockwise and the bead, B , is pulled along the wire towards Q .



- (b) Calculate BQ when angle $BAO = 110^\circ$. [3]

4 CCHY/2025/I/24

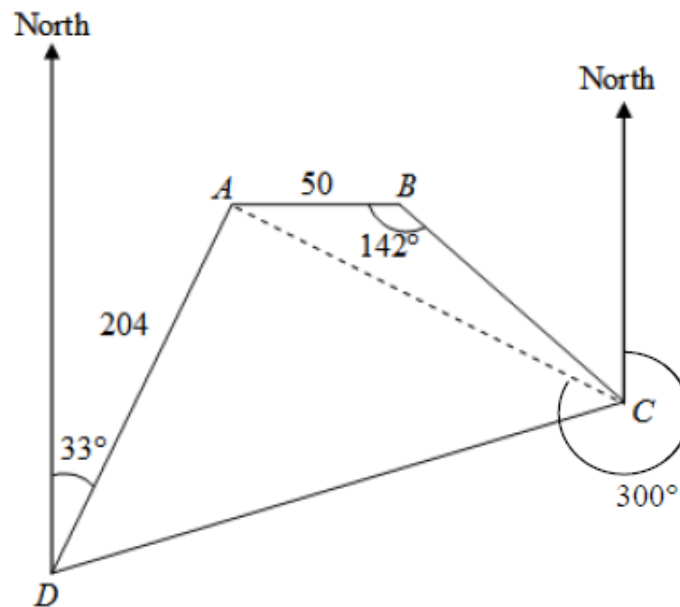
The diagram shown below is part of a model drawing of the neighborhood.
 Block B is due east of Block A . Block C has a bearing of 300° from B .
 The distance between Block B and Block C is 8 cm.

North



A ————— B

- (a) Mark and label the position of Block C on the model drawing. [1]
- (b) On the same diagram, construct the perpendicular bisector of the line AB . [1]
- (c) A letterbox is to be positioned such that it is equidistant from the three blocks.
 - (i) On the same diagram, mark L to represent the position of the letterbox. [2]
 - (ii) Given that the actual distance of $AB = 4$ km, calculate the actual distance between Block C and the letterbox, in kilometres. [1]



The diagram shows a plot of land with points $ABCD$ on level ground.
 The bearing of A from D is 033° and the length of AD is 204 m.
 B is 50 m due east of A .
 Angle $ABC = 142^\circ$ and the bearing of A from C is 300° .

- (a) Find the bearing of C from A .

Answer [1]

- (b) Find angle BAC .

Answer [1]

- (c) Show that the distance AC is 221.19 m, correct to two decimal places.

[2]

(d) Find the area of triangle ABC .

Answer m^2 [2]

(e) Find angle DAC .

Answer [1]

(f) Hence, calculate the distance DC .

[3]

(g) A tree, 15 m tall, stands vertically at A .

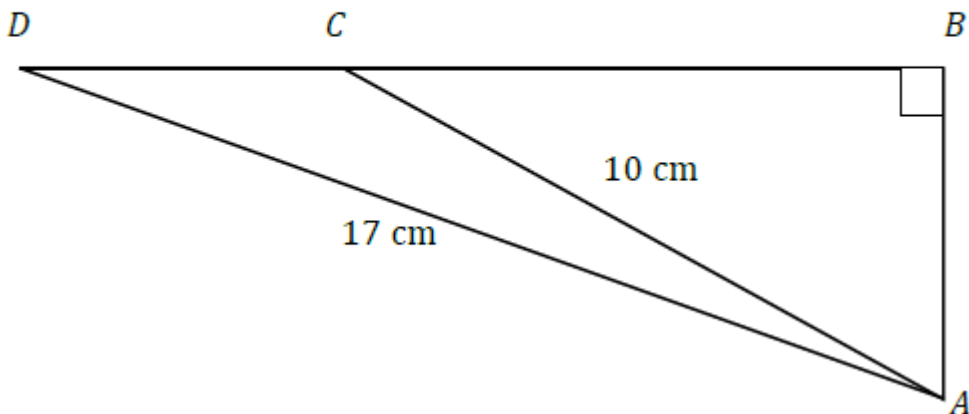
Find the angle of elevation of the top of the tree from D .

Answer [2]

6

SST/2025/I/15

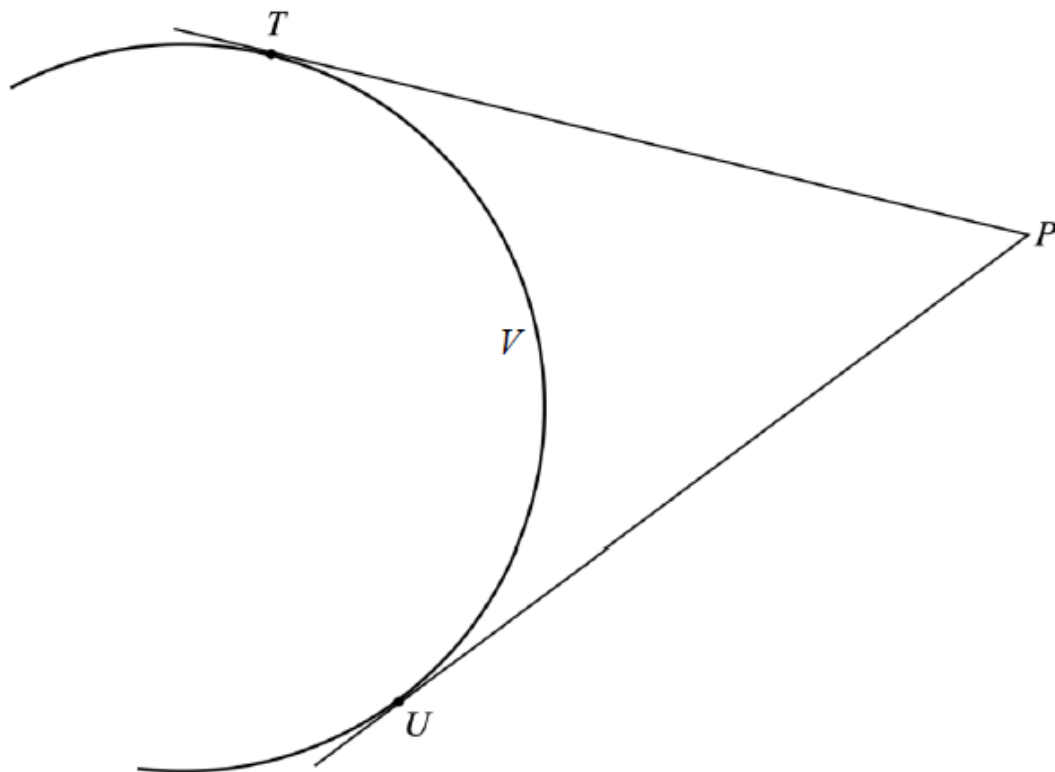
In the triangle ABD , $AD = 17$ cm, $AC = 10$ cm and $\angle ABC = 90^\circ$.



	(a) Given that $\sin \angle ACD = \frac{3}{5}$, find the exact value of (i) AB (ii) $\tan \angle ACB$ (b) A circle Q_1 is drawn such that it passes through A, B and C. A second circle Q_2 is drawn such that it passes through A, B and D. Find the ratio of the circumference of Q_1 to circumference of Q_2.
7	SST/2025/I/16
	Aris cycled from his house at Alley Way, point A, to meet Meng at point B, which is 500 m away on a bearing of 121°. After that, they both cycled to the School of Science and Technology (SST), point C, on a bearing of 342°, which is 410 m from B. (a) Show that angle ABC is 41°. (b) Calculate the distance AC. (c) Calculate the shortest distance from A to BC. (c) Calculate the shortest distance from A to BC.
8	SST/2025/I/23

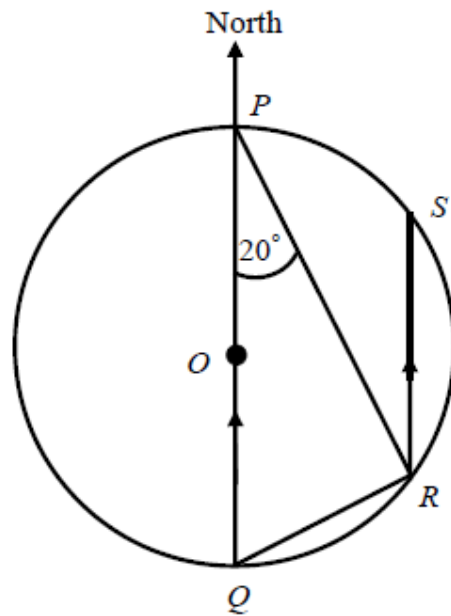
The diagram shows a plot of land, $PTVU$, on a campsite.

PT and PU are tangents of an arc of a circle at T and U .



- (a) Find the center of the circle and label it X . [1]
- (b) Construct the perpendicular bisector of TP . [1]
- (c) Construct the angle bisector of angle TPU . [1]
- (d) A lookout tower is to be built on the campsite $PTVU$, nearer to T than P and nearer to PT than PU .

Shade the region where the lookout tower is to be built. [1]



PQ is a diameter of the circle $PQRS$, with centre O .
 Angle $QPR = 20^\circ$, $PQ = 12$ km and QP is parallel to RS .

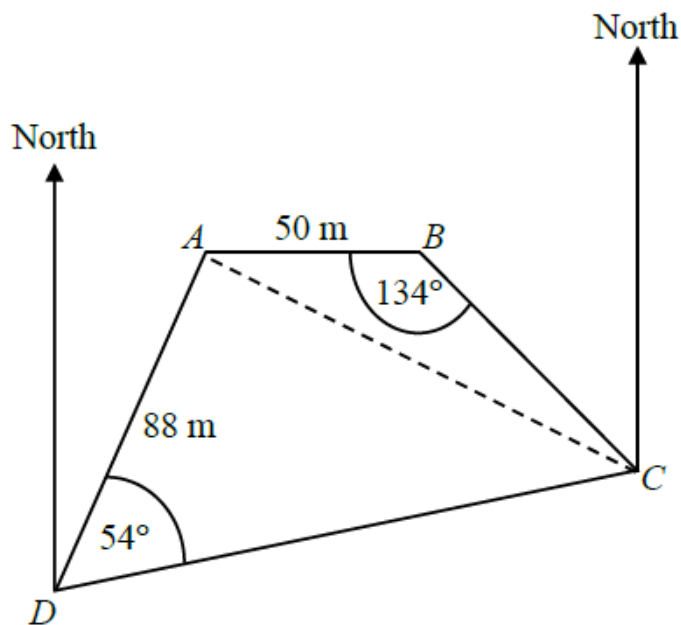
- (a) Explain why triangle PQR is a right-angled triangle.

.....
 [2]

- (b) P is due north of Q . Find the bearing of Q from R .

Answer [1]

- (c) Calculate the perimeter of triangle PQR . [2]



A farm with points A , B , C and D are on level ground.
 A is 88 m from D on a bearing of 033° .
 The bearing of A from C is 300° .
 Angle ABC is 134° and angle ADC is 54° .

- (a) B is 50 m due east of A .
 Show that angle BAC is 30° .

[2]

- (b) Calculate AC . [2]

(c) Calculate DC .

Answer $DC = \dots\dots\dots\text{m}$ [3]

(d) Find the shortest distance from point A to DC .

Answer $\dots\dots\dots\text{m}$ [2]

(e) Calculate the area of triangle ABC .

[2]

(f) A rat runs along the path DC .

An eagle sits on a branch 45 m directly above point A .

Find the smallest angle of depression of the rat from the eagle.

Answer $\dots\dots\dots$ [2]

(g) Explain why the answer to **part (f)** is an estimated value.

$\dots\dots\dots$

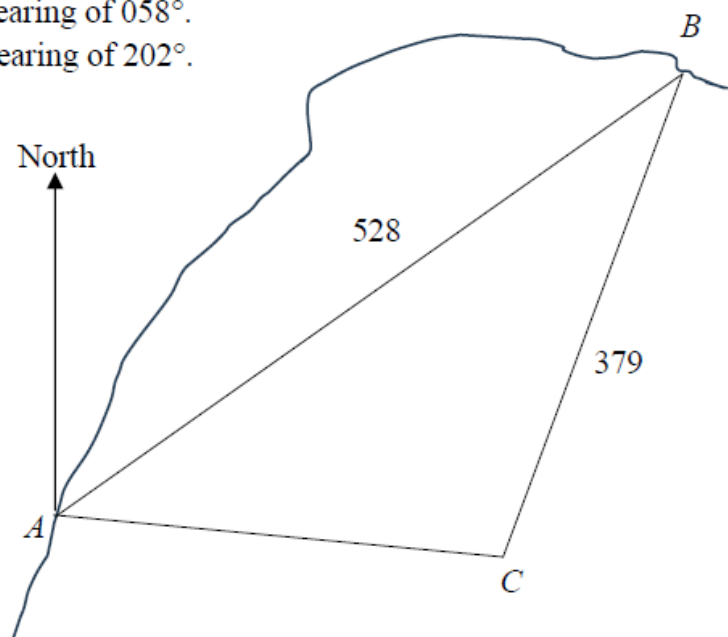
$\dots\dots\dots$ [1]

	(a) Given that $\cos x = -0.482$, find the value for angle x, where $0^\circ \leq x \leq 180^\circ$. <i>Answer</i> $x =$ _____ $^\circ$ [1] (b) The area of triangle ABC is 66.5 cm^2. $AB = 16.2 \text{ cm}$ and $BC = 13.5 \text{ cm}$. Find the smallest possible size of reflex angle ABC. [3]
12	ACSB/2025/II/05

Points A , B and C are at sea level, with A and B located on the shoreline.

B is 528 m from A on a bearing of 058° .

C is 379 m from B on a bearing of 202° .



(a) Calculate AC .

Answer _____ m [3]

(b) Calculate the bearing of A from C .

[3]

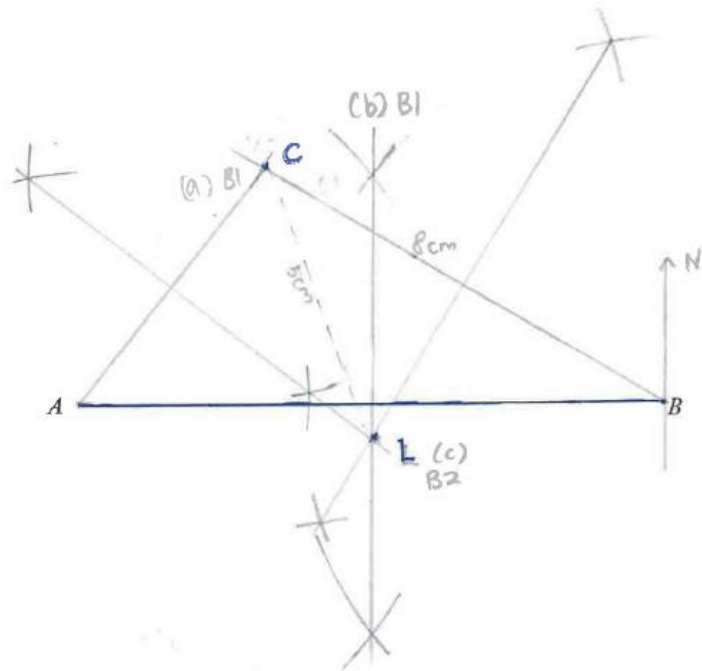
- (c) John paddles out to sea along AC when his paddleboard broke at a point nearest to B .
Find the distance between John and B when his paddleboard broke.

Answer _____m [2]

- (d) A lifeguard located on the watch tower at B spotted John.
Find the height of the watch tower if the greatest angle of depression of John from the lifeguard is 7° .

[2]

1	nil
2	(a) $560^2 = 520^2 + 331^2 - 2(520)(331)\cos(\angle CAB) \text{ --- [M1] correct value substituted}$ $\cos(\angle CAB) = \frac{520^2 + 331^2 - 560^2}{2(520)(331)} \text{ --- [M1]}$ $\angle CAB = 78.885 = 78.9^\circ \text{ --- [A1]}$ (b) $\text{Area} = \frac{1}{2}(331)(520)\sin(78.885) = 84445.7648 \text{ --- [M1]}$ Let shortest distance be x $\frac{1}{2}(560)x = 84445.7648 \text{ --- [M1] for area of triangle } \frac{1}{2}(560)x$ $x = 301.592$ Let height of building be H $\tan 18.4 = \frac{H}{301.592} \text{ --- [M1]}$ $H = 100m \text{ --- [A1]}$
3	(a) $\frac{\sin \hat{ABQ}}{20} = \frac{\sin 42^\circ}{60}$ $\hat{ABQ} = \sin^{-1} \left[\frac{\sin 42^\circ}{60} \times 20 \right]$ $\hat{ABQ} \approx 12.8878^\circ = 12.9^\circ \text{ (to 1 d.p.)}$ (b) $BO^2 = 60^2 + 20^2 - 2(60)(20)\cos 110^\circ$ $BO = \sqrt{60^2 + 20^2 - 2(60)(20)\cos 110^\circ}$ $BO \approx 69.4323$ $BQ \approx 69.4323 - 20$ $BQ \approx 49.4323 = 49.4 \text{ cm (to 3 s.f.)}$
4	(a) – (ci)



(cii)

2 km

5

(a)

$$\begin{aligned}\text{Bearing of } C \text{ from } A &= 180^\circ - (360^\circ - 300^\circ) \\ &= 120^\circ\end{aligned}$$

(b)

$$\begin{aligned}\angle BAC &= 120^\circ - 90^\circ \\ &= 30^\circ\end{aligned}$$

(c)

$$\frac{AC}{\sin 142^\circ} = \frac{50}{\sin 8^\circ}$$

$$\begin{aligned} AC &= \frac{50 \sin 142^\circ}{\sin 8^\circ} \\ &\approx 221.1855 \\ &\approx 221.19 \text{ m (Shown)} \end{aligned}$$

(d)

$$\begin{aligned}\text{Area of triangle } ABC &= \frac{1}{2} \times 50 \times 221.19 \sin 30^\circ \\ &= 2764.875 \\ &\approx 2760 \text{ m}^2\end{aligned}$$

(e)

$$\begin{aligned}\angle DAC &= 33^\circ + 60^\circ \\ &= 93^\circ\end{aligned}$$

(f)

$$DC^2 = 204^2 + 221.19^2 - 2 \times 202 \times 221.19 \cos 93^\circ$$

$$DC = \sqrt{95264.10}$$

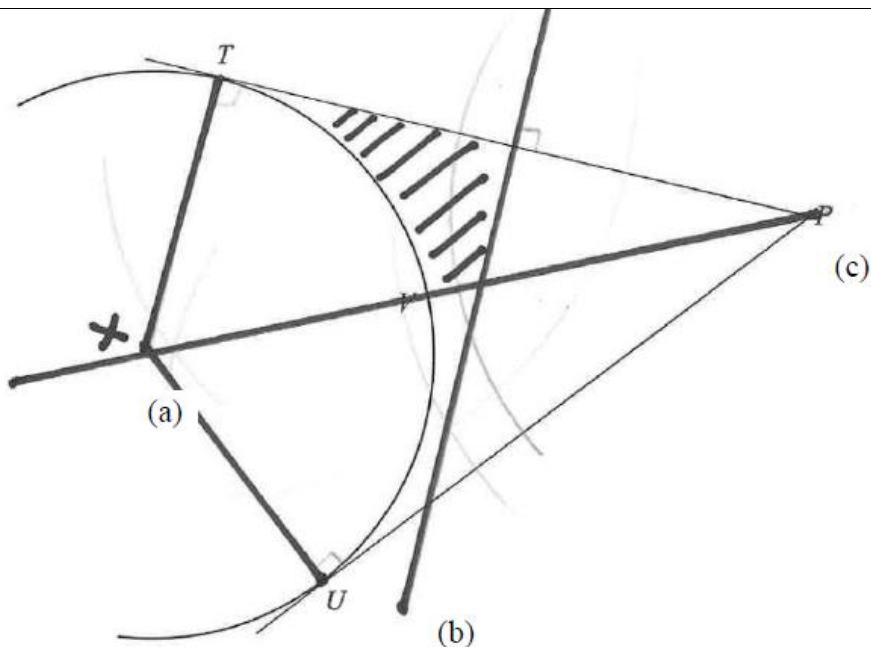
≈ 308.6488

$\approx 309 \text{ m}$

(g)

	Let angle of elevation be θ. $\tan \theta = \frac{15}{204}$ $\theta = \tan^{-1}\left(\frac{15}{204}\right)$ ≈ 4.20536 $\approx 4.2^\circ \text{ (to 1 d.p.)}$
6	(ai) $\sin \angle ACD = \sin \angle ACB = \frac{3}{5}$ $\frac{AB}{AC} = \frac{AB}{10} = \frac{3}{5}$ $AB = 6 \text{ cm}$ (aii) $BC = \sqrt{10^2 - 6^2} = 8 \text{ cm}$ $\tan \angle ACB = \frac{AB}{BC} = \frac{6}{8} = \frac{3}{4}$ (b) Note that AC is the diameter of Q_1 (right angle in a semicircle) And AD is the diameter of Q_2 (right angle in a semicircle) Therefore $\frac{\text{circumference of } Q_1}{\text{circumference of } Q_2} = \frac{10}{17}$ The ratio of the circumference of Q_1 to circumference of Q_2 is 10 : 17.
7	(a) $\angle ABN = 180^\circ - 121^\circ = 59^\circ \text{ (interior angles, parallel lines)}$ $\angle CBN = 360^\circ - 342^\circ = 18^\circ \text{ (angles at a point)}$ $\therefore \angle ABC = 59^\circ - 18^\circ = 41^\circ \text{ (adjacent angles) (shown)}$ (b) $AC^2 = AB^2 + BC^2 - 2(AB)(BC)\cos 41^\circ$ $AC^2 = 500^2 + 410^2 - 2(500)(410)\cos 41^\circ$ $AC = 329.64 = 330 \text{ m (correct to 3 significant figures)}$ (c) Let the shortest distance from A to BC be h. $\sin \angle ABC = \frac{h}{500}$ $h = 500 \sin 41^\circ$ $h = 328.02 = 328 \text{ m (correct to 3 significant figures)}$

8



9

(a)

Angle $PRQ = 90^\circ$ and by the property right angle in semicircle.
Therefore, triangle POR is a right-angled triangle.

(b)

angle $QRS = 90^\circ + 20^\circ = 110^\circ$ (alt angles, PQ parallel to SR)
 Bearing $= 360^\circ - 110^\circ = 250^\circ$

(c)

$$\begin{aligned}\text{Perimeter} &= 12 + 12\sin 20^\circ + 12 \cos 20^\circ \\ &= 27.4 \text{ km}\end{aligned}$$

10

6a

Bearing of C from A

$$= 180^\circ - 60^\circ$$

$$= 120^\circ$$

Angle BAC

$$= 120^\circ - 90^\circ$$

$$= 30^\circ$$

6b

Angle BCA

$$= 180^\circ - 134^\circ - 30^\circ$$

$$= 16$$

$$\frac{AC}{\sin B} = \frac{AB}{\sin C}$$

$$AC = \frac{50 \sin 134^\circ}{\sin 16^\circ}$$

$$= 130.49$$

$$= 130 \text{ m (correct to 3 sf)}$$

	6c	Angle $DAC = 60^\circ + 33^\circ = 93^\circ$	
		<u>Method 1:</u> Using cosine rule, $DC^2 = DA^2 + AC^2 - 2(DA)(AC)\cos DAC$ $= 88^2 + (130.49)^2 - 2(88)(130.49)\cos 93^\circ$ $DC = 161.163272$ $= 161 \text{ m (correct to 3 sf)}$	<u>Method 2:</u> Using sine rule, $\frac{DC}{\sin 93^\circ} = \frac{88}{\sin 33^\circ}$ $DC = \frac{88\sin 93^\circ}{\sin 33^\circ}$ $= 161.3534716$ $= 161 \text{ m (correct to 3sf)}$
	6d	Let d be the shortest distance $\sin 54^\circ = \frac{d}{88}$ $d = 88\sin 54^\circ$ $= 71.193$ $= 71.2 \text{ m (correct to 3sf)}$	
	6e	Area of triangle ABC $= \frac{1}{2}(130.49)(50)\sin 30^\circ$ $= 1631.125$ $= 1630 \text{ m}^2 \text{ (correct to 3sf)}$	
	6f	$\tan \theta = \frac{45}{130.49}$ $\theta = 19.027^\circ$ $= 19.0^\circ \text{ (correct to 1 dp)}$	
	6g	- The height 45 m excludes the vertical distance between the eagle's eyes from the branch. - The rat might not run in a straight line in reality. - The length of DC (130.49 m) is an estimation.	
11	(a)	118.8	
	(b)	$\frac{1}{2}(16.2)(13.5)\sin \angle ABC = 66.5$	
		$\angle ABC = 37.455^\circ \text{ or } 142.54^\circ$	
		Smallest possible size = $360 - 142.54$ $= 217.5$	
12	(a)	$\angle ABC = 360 - 202 - (180 - 58) = 36^\circ$	

[illegible]